CHAPTER 5 · CONCEPT 1 · WORKSHEET

Patterns, Parity & Multiples — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ **[1]**

Take four consecutive numbers and place a '+' or '−' sign between them. How many different expressions can you form?

(a) 4 (b) 6 (c) 8 (d) 16

Q2 · MCQ **[1]**

Which of these is always even for every integer n?

(a) $2n + 1$ (b) $3n$ (c) $4n$ (d) $n + 1$

Q3 · FILL IN THE BLANK **[1]**

The sum of three consecutive even numbers is 72. The middle number is _______.

____

Q4 · MCQ **[1]**

How many multiples of 7 are there between 200 and 300?

(a) 13 (b) 14 (c) 11 (d) 15

Q5 · TRUE / FALSE **[1]**

$n(n + 1)$ is always even for any integer n.

True False

Q6 · ASSERTION–REASON [1]

Assertion (A): A number divisible by 9 is also divisible by 3.
Reason (R): 9 is a multiple of 3.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7 [2]

The greatest of seven consecutive integers is k. Write the other six numbers in terms of k.

Q8 [2]

Show, using algebra, that 2a + 2b is always even.

Q9 [2]

If a number is divisible by both 9 and 4, what is the smallest number it must also be divisible by? Explain.

Long answer (3 marks each)

Q10 [3]

The sum of four consecutive numbers is 34. Find the numbers, showing your method.

Q11 **[3]**

For each, state whether it is always / sometimes / never even, and justify with algebra: (i) $4x + 6$ (ii) $3g + 5h$.

Q12 **[3]**

Prove that the sum of four consecutive integers is always even.

Total: 21 marks · Check your work against the answer key.